Computer Forensics Case Studies

Case Study 5 Investigating DJI Mavic Air Flight Evidence

HUSSEINI SULEIMAN

2025/FWSD/11478

fwsd2511478@students.icdfa.edu.ng

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Student Assessment Brief

Assessment detail	Information
Course	SBT-DF204 Computer Forensics Case Studies
Assessment	Case Study 5 Individual mobile and microSD forensic investigation
Case title	Investigating DJI Mavic Air Flight Evidence
Submission format	Written report with labelled screenshots, timeline, and evidence log
Deadline and maximum score	As published in E-Campus

This individual case study asks you to examine DJI Mavic Air evidence from an Android mobile extraction, a raw internal microSD image, and an Expert Witness Format image. You will identify and compare DJI GO 4 flight records, cached video, detailed records, cached pictures, and media artefacts. Your report must build an evidence-led flight timeline while clearly separating direct observations from inferences about a flight, location, device, or operator.

Case scenario

Investigators recovered an Android logical acquisition associated with the DJI GO 4 application and two representations of DJI Mavic Air internal microSD media. The enquiry is whether the sources support a reliable account of aircraft activity, recorded media, and the relationship between the mobile application and microSD evidence. The case material may contain flight locations, images, and records relating to people. This is an offline examination of authorised course artefacts. It does not authorise you to fly, power on, connect to, synchronise, reset, or change settings on any drone, mobile device, account, cloud service, or removable media.

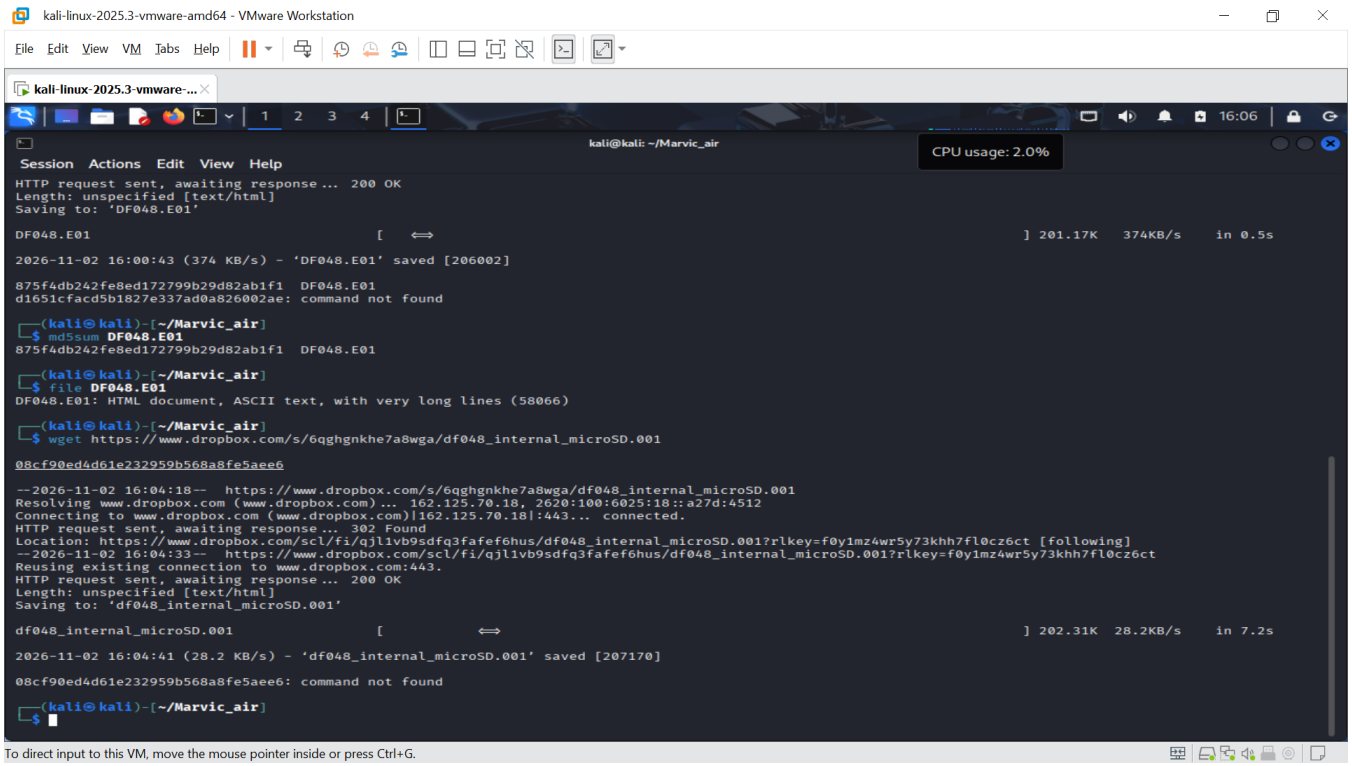
Case materials

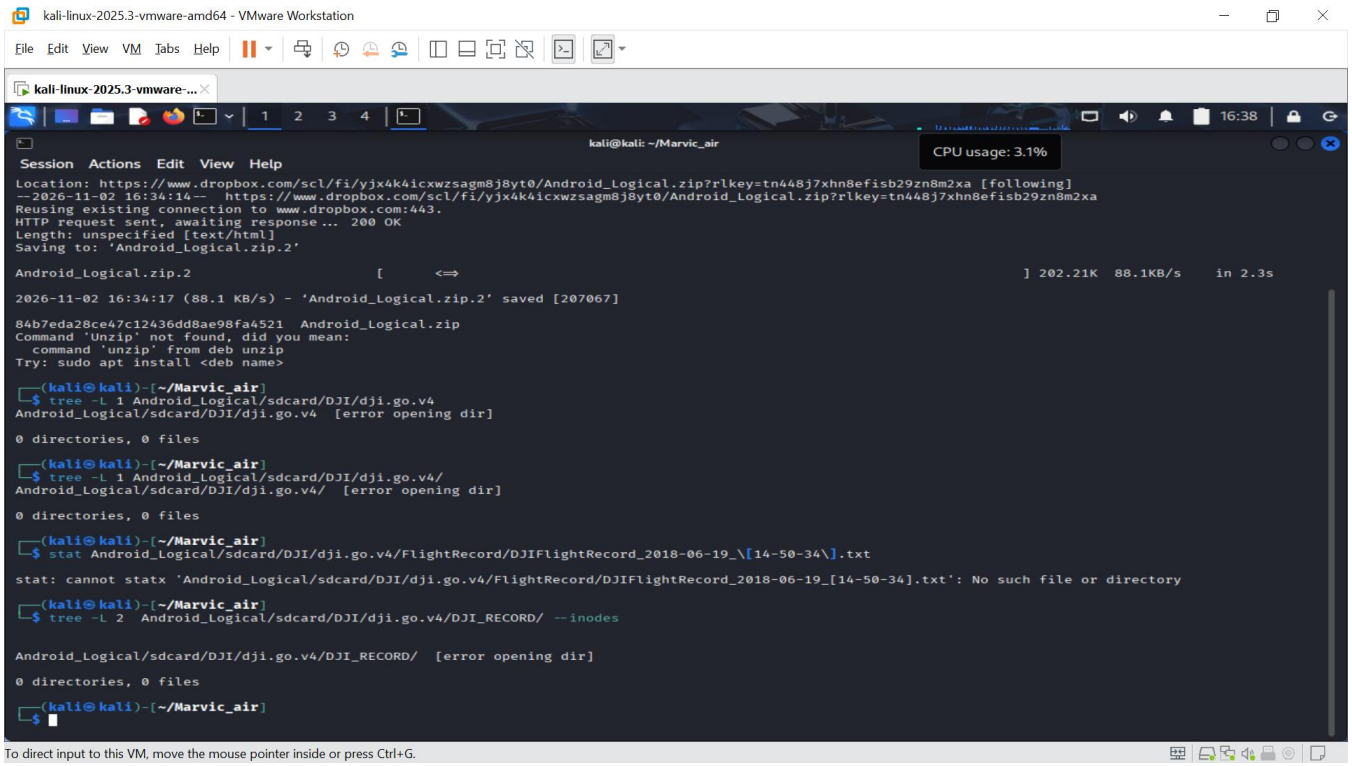
Use only the Android logical image and report, the raw Mavic internal microSD image, the EWF image, and accompanying teaching decks supplied through E-Campus. The DJI GO 4 application package is dji.go.v4. The mobile deck introduces four artefact families: main flight records in TXT format, cached MP4 video,

detailed DAT flight records, and cached pictures. The microSD decks cover raw and EWF examination. Use the exact filenames supplied in your package, create working copies, and record the source and integrity value of each original before analysis.

What you must submit

Item	Required output	What to provide
1	Forensic working set	Recors The Android logical image: **Android_Logical.zip.1** mobile forensic report: **84b7eda28ce47c12436dd8ae98fa4521** Android_Logical.zip Raw microSD image, EWF image by original file name, Byte size: **207067 bytes** SHA-256 value: **ab32c8d81417b554d15a1b03b9d88a2105f4acad8611e7564d8f03167b9ddb06** Android_Logical.zip.1 source location: working-copy location.: 529c224681cf2b75d464b9734c1cc9a1 df048_internal_microSD.001.1Preserve every original unchanged.: 875f4db242fe8ed172799b29d82ab1f1
2	DJI GO 4 evidence record	Provide labelled screenshots or saved output for the dji.go.v4 artefacts: main flight records, cached video, detailed flight records, cached pictures, associated metadata, and your time-zone treatment.
3	Media comparison record	Document the raw microSD and EWF image formats, partitions or file listings, relevant media hashes and timestamps, microSD identifier or image serial information, and cross-source comparison.
4	Student report	Submit one report in PDF or DOCX format with a completed evidence log and a UTC flight timeline. Include only the supporting files requested by your lecturer.





Investigation task

Conduct the investigation in stages: preserve the supplied sources; inventory DJI GO 4 artefacts; interpret each record type according to its event scope; examine the raw microSD image; examine the EWF image; compare the material sources; then create one UTC timeline. A main flight record, video cache, and detailed

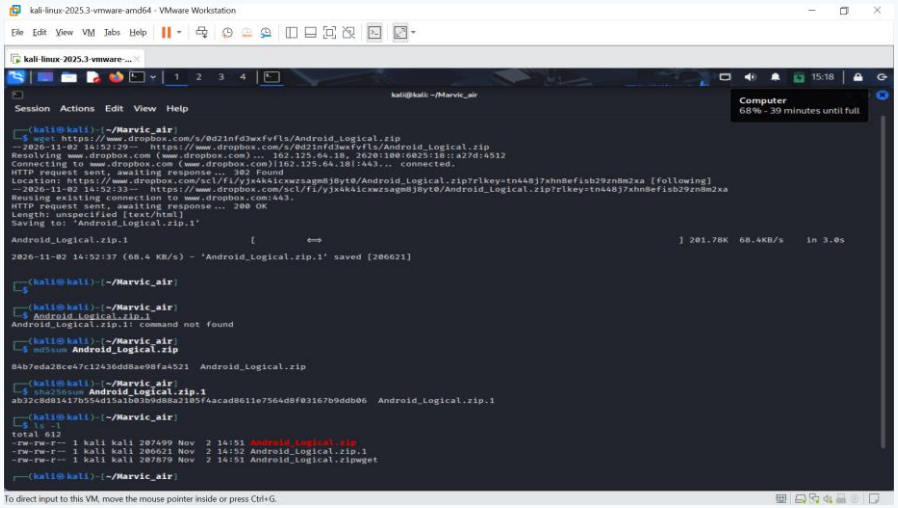
record use different start and end conditions, so you must not assume that their timestamps describe the same event. Base each finding on identifiable data such as a file path, hash, parser output, media identifier, image serial, timestamp, metadata field, or screenshot.

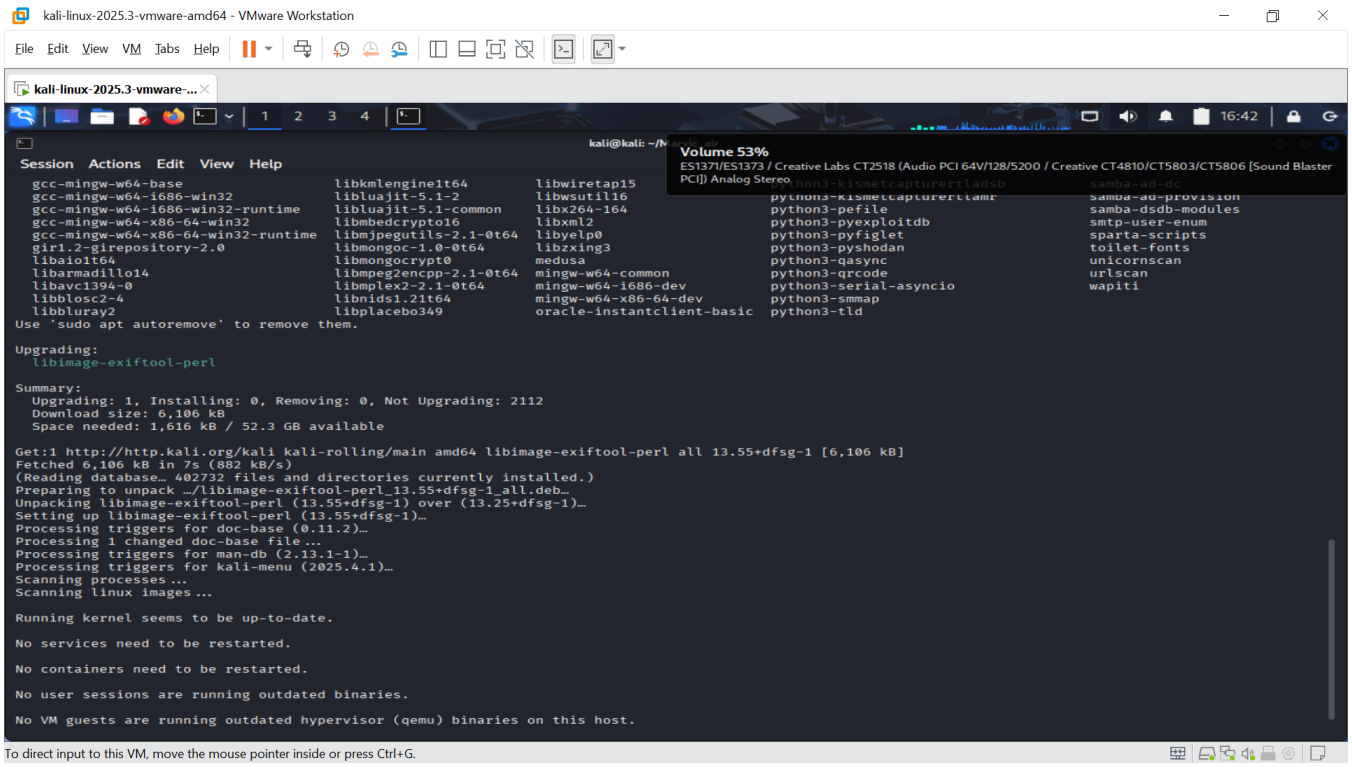
No	Investigation question	Minimum evidence to discuss
1	How did you preserve the case materials?	For each mobile and microSD source, record the source, original file name, byte size, SHA-256 value, working-copy method, tool version, and read-only or access precautions.
2	What DJI GO 4 artefacts are present?	Locate and document the DJI GO 4 application data and identify the folders or paths used for main flight records, cached video, detailed flight records, and cached pictures.
3	What does the main flight record establish?	Use the supplied course workflow to interpret the relevant TXT record. Identify device or aircraft information, start and end events, location evidence, duration, source time, UTC conversion, and any time-zone anomaly.
4	What does the cached video establish?	Identify the MP4 source path, filename, start and end information, duration, metadata, hash, and its relationship to the main flight record. Preserve metadata before previewing a working copy.
5	What does the detailed flight record establish?	Examine the DAT record using the approved course workflow. Explain its power-on to power-off scope and how it corroborates or differs from motor-on, take-off, landing, and video-recording evidence.
6	What do cached pictures establish?	Record paths, file types, capture sequence, timestamps, hashes, and the limits of using visual content or cache artefacts to infer a flight, location, or operator.
7	What does the raw internal microSD image show?	Document image format, integrity, sector or partition layout, read-only mount method, relevant videos and pictures, file metadata, hashes, and the microSD identifier if available.
8	What does the EWF image show and how does it compare?	Verify format and integrity, examine the mounted EWF data, record the image serial or media identifier, and compare media hashes, timestamps, and other artefacts with the raw image.
9	What conclusion can you defend?	Build a UTC flight timeline, distinguish direct observations from inference, state a confidence level, and explain at least three limitations or alternative explanations about device ownership, operator identity, route, and time.

Technical starting points

The following are course-aligned starting points. They do not replace documentation of your own method. Work only on copies, keep the original sources unchanged, and preserve the output that supports each decision.

Purpose	Course-aligned starting point
Preserve and identify evidence	Create read-only originals and working copies. Record SHA-256 values before and after copying; record the supplied mobile report as a reference source, not as a substitute for your own verification.
Locate DJI GO 4 artefacts	Use the Android logical extraction and the course-provided mobile report to locate dji.go.v4 data. Record full paths for TXT, MP4, DAT, and cached-picture artefacts.
Interpret main flight records	Use the approved offline parser or manual workflow demonstrated in class. Record the input file, tool or parser, version, source time, UTC conversion, and any ambiguity in the displayed time zone.

Purpose	Course-aligned starting point
Validate video and pictures	Record file type, byte size, filename, timestamps, and SHA-256 value before opening a working copy. Use metadata output and a controlled preview, then preserve the supporting screenshot. 
Examine raw microSD media	Use a read-only mount or forensic tool workflow. Document image type, sector or partition structure, file listings, timestamps, hashes, and relevant media identifiers.
Examine EWF media	Use the approved EWF workflow, such as ewfmount <image> <mount_point>, then inspect the exposed data read-only. Record the image format, integrity result, serial or GUID information, and access permissions.
Corroborate sources	Compare hashes, timestamps, file names, media identifiers, record scopes, and content only after normalising time. Explain both matches and differences between the mobile, raw microSD, and EWF sources.



Evidence interpretation safeguards

- Treat each record type according to its stated scope. A record beginning at power on, motor on, take-off, video start, landing, or power off supports a different part of the sequence.
- Document the source time zone, UTC conversion, and any apparent device time-setting problem. Do not use a time in a filename as proof without validating it against record content and other artefacts.
- A file hash can establish binary equality between copies, but it does not alone prove when a flight occurred, who operated the aircraft, or where it was flown.
- Use a read-only workflow for the microSD images. Record any mount location, permission or ownership change, tool version, partition view, and output path so another examiner can repeat your steps.
- Handle cached pictures, video, GPS information, and records about people as sensitive evidence. Redact unrelated personal or location data in screenshots while retaining enough forensic detail for assessment.
- Do not upload case files to external drone portals, flight parsers, cloud accounts, public scanners, or media-sharing services. Do not connect the evidence to live DJI equipment or execute any untrusted artefact.

Required report structure

1. Case reference and student identification. Include your full name, registration number, course code, and case title.
2. Executive conclusion. State a concise, cautious conclusion and confidence level. Do not present a person-level attribution unless the evidence supports it.
3. Evidence acquisition and integrity. Record sources, filenames, sizes, hashes, originals, working copies, tools, versions, and access controls.
4. Method. Explain the mobile-artifact, media-metadata, raw microSD, EWF, comparison, and time-normalisation methods used.
5. Findings. Answer every investigation question with a precise evidence reference: path, hash, parser output, timestamp, media identifier, image serial, metadata field, or screenshot ID.
6. Timeline and correlation. Present the material sequence in UTC and explain how TXT, MP4, DAT, cached pictures, raw microSD, and EWF evidence corroborate or conflict.

7. Conclusion and limitations. Explain gaps, altered access times, cache artefact limits, time-zone uncertainty, partial media, app versus device ownership, and the difference between technical records and proof of a person at the controls.
8. Appendices. Include labelled screenshots, selected tool output, the completed timeline, completed evidence log, and only the supporting material requested by your lecturer.

Timeline template

Complete a version of this table in your report or evidence appendix. Add rows for all material events and keep the main timeline in UTC.

Stage	UTC timestamp	Evidence ID	Source artefact	Observed fact	Interpretation and limit
Device state and power or motor on					
Take off or video start					
Main flight record event					
Landing or video end					
Power off and correlation					

Evidence log template

Use an evidence log that lets an assessor locate the exact source file, path, parser, mount method, hash, timestamp, media identifier, image serial, and screenshot that supports each finding.

Evidence ID	Tool and method	Source record, path, or image	Observed fact	Screenshot or appendix reference
E01				
E02				
E03				
E04				
E05				
E06				
E07				
E08				

Submission rules

- Work individually. Your analysis, report wording, screenshots, timeline, and evidence log must be your own work.
- Submit the report as SBTDF204_CaseStudy5_RegistrationNumber.pdf or .docx. If separate supporting material is requested, use SBTDF204_CaseStudy5_Evidence_RegistrationNumber.zip.

- Do not upload the original Android image, mobile report, raw microSD image, EWF image, recovered media, or source archive unless your lecturer specifically requests it. Submit only the report and required supporting evidence.
- Use only the authorised, offline course artefacts in the approved lab environment. Do not connect to drones, controllers, mobile accounts, cloud services, or external flight-analysis portals. Do not change evidence settings or publicly upload or share case evidence.
- Redact unrelated personal or sensitive data in screenshots while retaining enough path, timestamp, hash, parser, mount, and tool detail for the examiner to verify your finding.

Marking guide

The E-Campus maximum score applies to this case study. The assessor will apply the following percentage weighting to that maximum score.

Criterion	What strong work demonstrates	Weight
Evidence acquisition and integrity	Preserves each supplied source, records integrity and working copies, and explains the evidence-handling method accurately.	15%
DJI GO 4 mobile artefact analysis	Locates and interprets the TXT, MP4, DAT, and cached-picture artefacts with evidence references, time-zone treatment, and defensible limitations.	20%
Raw microSD image analysis	Examines the raw image read-only, documents layout and media artefacts, validates files, and records relevant metadata and identifiers.	20%
EWF image validation and comparison	Verifies the EWF image, records its media information, and compares it carefully with the raw microSD and mobile findings.	20%
Timeline, correlation, conclusion, and limitations	Builds a coherent UTC timeline, explains cross-source corroboration, and reaches a cautious conclusion about flight activity and attribution.	15%
Report quality and evidence presentation	Writes clearly, labels screenshots and output, cites evidence IDs consistently, and supplies a usable evidence log.	10%

Final check before submission

- Every major conclusion cites a source artefact, path, method, hash or media identifier, timestamp, and screenshot or appendix reference.
- Your timeline distinguishes the boundaries of power, motor, take-off, video recording, landing, and power-off evidence and uses UTC consistently.
- Your report explains how the mobile, raw microSD, and EWF sources corroborate or challenge the proposed flight sequence.
- Your conclusion distinguishes evidence of a device, application, route, image, video, or account from proof of a particular operator or person.